10 — Cash Flow & Runway

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> *All assumptions, rates, scenarios, and phase definitions in this document are sourced from params.md. Refer to that document for any parameter clarification.*

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1. Executive Summary

This document projects Health Hub's monthly cash position from Month 0 through Month 34 (the maximum timeline per params.md Section 13) across all 21 scenario combinations. The analysis answers the fundamental survival question: **when does the money run out for each path, and what can be done about it?**

Key findings:

- **S1-L1 (bootstrapped, minimal spend):** Lowest absolute burn (~$8,700/mo in early phases) but the slowest revenue ramp. Cash-out risk at Month 10-12 with $80K starting capital. Requires $105-$130K total to reach Production.
- **S1-L2 (self-funded, ideal spend):** The recommended self-funded baseline. Burns ~$14,900-$19,600/mo by phase. Requires ~$200-$260K in founder capital to reach Production without external investment. Cash-out risk at Month 12-14 with $200K starting capital.
- **S2-O2-I2 (ideal own spend, ideal investor at transition):** The recommended balanced path. Self-funded burn of ~$14,900-$17,600/mo through Pilot 1, then investor capital of $50-$80K extends runway through Pilot 2 and into Production. Cash-out risk effectively eliminated if investor arrives on schedule.
- **S3-O2-I2 (ideal own spend, ideal investor after Pilot 1):** Earlier investor arrival reduces self-funded requirement to ~$60-$90K. Investor provides $50-$80K at Month 5-8. Combined runway reaches 20+ months.
- **Across all 21 paths:** The critical danger zone is Month 8-14 for self-funded paths, when Pilot 1 ops costs begin but revenue has not yet materialized. Every path that does not secure external capital or achieve early revenue faces a cash crunch in this window.

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2. Monthly Cash Flow Model

2.1 Cost Assumptions (from params.md)

Cost Component	Pre-Pilot	Pilot 1	Pilot 2	Production
Team cost	$14,830/mo	$14,830/mo	$14,830/mo	$14,830/mo
Ops staff	$0/mo	$2,482/mo	$4,361/mo	$6,542/mo
Infra (midpoint)	$65/mo	$270/mo	$460/mo	$800/mo
Total outflow	**$14,895/mo**	**$17,582/mo**	**$19,651/mo**	**$22,172/mo**

Notes:

- Team cost is fixed at $14,830/mo across all phases (per params.md Section 3.2)
- Ops staff ramps per params.md Section 4.2: 0 in Pre-Pilot, then 2 doctors + 2 admin + 2 tech support in Pilot 1, scaling up
- Infra uses midpoint of ranges from params.md Section 7.4
- One-time equipment costs (phones $200-$300 in Pre-Pilot, laptops $400-$600 in Pilot 1, expansion $300-$500 in Pilot 2 per params.md Section 7.3) are included in the month they occur

Investment level adjustments:

- L1 (Bootstrapped): 40% of ideal = costs multiplied by 0.40 for team; ops and infra at minimum of ranges
- L2 (Ideal): 100% of baseline = costs as shown above
- L3 (Fully Funded): 150-200% of ideal = costs multiplied by 1.50-1.75 for team; ops and infra at maximum of ranges

2.2 Revenue Assumptions (from params.md Section 11.3)

Phase	Monthly Revenue Range	Midpoint Used in Model
Pre-Pilot	$0	$0
Pilot 1	$0-$500	$150
Pilot 2	$500-$3,000	$1,500
Production	$3,000-$15,000	$7,500

2.3 Path A: S1-L2 (Self-Funded, Ideal Spend)

Starting capital: $200,000 (founder-provided) **Phase durations:** Pre-Pilot 4 months, Pilot 1 3 months, Pilot 2 6 months, Production 8 months = 21 months total

Month	Phase	Team	Ops Staff	Infra	One-Time	Total Outflow	Revenue	Net Cash Flow	Cumulative Cash

1	Pre-Pilot	$14,830	$0	$65	$250	$15,145	$0	-$15,145	$184,855
2	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	-$14,895	$169,960
3	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	-$14,895	$155,065
4	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	-$14,895	$140,170
5	Pilot 1	$14,830	$2,482	$270	$500	$18,082	$50	-$18,032	$122,138
6	Pilot 1	$14,830	$2,482	$270	$0	$17,582	$100	-$17,482	$104,656
7	Pilot 1	$14,830	$2,482	$270	$0	$17,582	$200	-$17,382	$87,274
8	Pilot 2	$14,830	$4,361	$460	$400	$20,051	$500	-$19,551	$67,723
9	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$800	-$18,851	$48,872
10	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,100	-$18,551	$30,321
11	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,500	-$18,151	$12,170
12	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,800	-$17,851	**-$5,681**
13	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$2,200	-$17,451	-$23,132
14	Production	$14,830	$6,542	$800	$0	$22,172	$3,000	-$19,172	-$42,304
15	Production	$14,830	$6,542	$800	$0	$22,172	$4,000	-$18,172	-$60,476
16	Production	$14,830	$6,542	$800	$0	$22,172	$5,500	-$16,672	-$77,148
17	Production	$14,830	$6,542	$800	$0	$22,172	$7,000	-$15,172	-$92,320
18	Production	$14,830	$6,542	$800	$0	$22,172	$8,500	-$13,672	-$105,992
19	Production	$14,830	$6,542	$800	$0	$22,172	$10,000	-$12,172	-$118,164
20	Production	$14,830	$6,542	$800	$0	$22,172	$12,000	-$10,172	-$128,336
21	Production	$14,830	$6,542	$800	$0	$22,172	$14,000	-$8,172	-$136,508

Cash-out month: Month 12. S1-L2 with $200K starting capital runs out of cash during Pilot 2 unless additional capital is injected or revenue accelerates beyond projections.

Total capital required for 21 months: ~$336,500 ($200K starting + ~$136.5K shortfall)

2.4 Path B: S2-O2-I2 (Ideal Own Spend, Ideal Investor at Transition)

Starting capital: $100,000 (founder-provided) **Investor capital:** $75,000 arriving at Month 8 (Pilot 1 to Pilot 2 transition) **Phase durations:** Pre-Pilot 4 months, Pilot 1 3 months, Pilot 2 5 months, Production 8 months = 20 months total

Month	Phase	Team	Ops Staff	Infra	One-Time	Total Outflow	Revenue	Investor	Net Cash Flow	Cumulative Cash
1	Pre-Pilot	$14,830	$0	$65	$250	$15,145	$0	$0	-$15,145	$84,855
2	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	$0	-$14,895	$69,960
3	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	$0	-$14,895	$55,065
4	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	$0	-$14,895	$40,170
5	Pilot 1	$14,830	$2,482	$270	$500	$18,082	$50	$0	-$18,032	$22,138
6	Pilot 1	$14,830	$2,482	$270	$0	$17,582	$100	$0	-$17,482	$4,656
7	Pilot 1	$14,830	$2,482	$270	$0	$17,582	$200	$0	-$17,382	-$12,726
8	Pilot 2	$14,830	$4,361	$460	$400	$20,051	$500	**$75,000**	+$55,449	$42,723
9	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$800	$0	-$18,851	$23,872
10	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,200	$0	-$18,451	$5,421
11	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,600	$0	-$18,051	-$12,630
12	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$2,000	$0	-$17,651	-$30,281
13	Production	$14,830	$6,542	$800	$0	$22,172	$3,500	$0	-$18,672	-$48,953
14	Production	$14,830	$6,542	$800	$0	$22,172	$5,000	$0	-$17,172	-$66,125
15	Production	$14,830	$6,542	$800	$0	$22,172	$7,000	$0	-$15,172	-$81,297
16	Production	$14,830	$6,542	$800	$0	$22,172	$9,000	$0	-$13,172	-$94,469
17	Production	$14,830	$6,542	$800	$0	$22,172	$11,000	$0	-$11,172	-$105,641
18	Production	$14,830	$6,542	$800	$0	$22,172	$13,000	$0	-$9,172	-$114,813
19	Production	$14,830	$6,542	$800	$0	$22,172	$14,500	$0	-$7,672	-$122,485
20	Production	$14,830	$6,542	$800	$0	$22,172	$15,000	$0	-$7,172	-$129,657

Cash-out month: Month 7 (briefly negative before investor arrival at Month 8). This path requires either a bridge arrangement for the Month 7 gap or earlier investor commitment. With $125K founder capital instead of $100K, the gap disappears.

With $125K founder capital: Cash-out shifts to Month 11 (post-investor). Total additional capital needed: ~$129.7K beyond the $100K founder + $75K investor = $54.7K shortfall.

Realistic assessment: S2-O2-I2 requires ~$100-$125K founder capital + $75-$100K investor capital + revenue acceleration or a second smaller raise to sustain through Production.

2.5 Path C: S3-O2-I2 (Ideal Own Spend, Ideal Investor After Pilot 1)

Starting capital: $75,000 (founder-provided) **Investor capital:** $75,000 arriving at Month 5 (after Pilot 1 begins, earlier than S2) **Phase durations:** Pre-Pilot 3 months, Pilot 1 2 months, Pilot 2 5 months, Production 8 months = 18 months total

Month	Phase	Team	Ops Staff	Infra	One-Time	Total Outflow	Revenue	Investor	Net Cash Flow	Cumulative Cash
1	Pre-Pilot	$14,830	$0	$65	$250	$15,145	$0	$0	-$15,145	$59,855
2	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	$0	-$14,895	$44,960
3	Pre-Pilot	$14,830	$0	$65	$0	$14,895	$0	$0	-$14,895	$30,065
4	Pilot 1	$14,830	$2,482	$270	$500	$18,082	$50	$0	-$18,032	$12,033
5	Pilot 1	$14,830	$2,482	$270	$0	$17,582	$150	**$75,000**	+$57,568	$69,601
6	Pilot 2	$14,830	$4,361	$460	$400	$20,051	$500	$0	-$19,551	$50,050
7	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$900	$0	-$18,751	$31,299
8	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,300	$0	-$18,351	$12,948
9	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$1,800	$0	-$17,851	-$4,903
10	Pilot 2	$14,830	$4,361	$460	$0	$19,651	$2,500	$0	-$17,151	-$22,054
11	Production	$14,830	$6,542	$800	$0	$22,172	$4,000	$0	-$18,172	-$40,226
12	Production	$14,830	$6,542	$800	$0	$22,172	$6,000	$0	-$16,172	-$56,398
13	Production	$14,830	$6,542	$800	$0	$22,172	$8,000	$0	-$14,172	-$70,570
14	Production	$14,830	$6,542	$800	$0	$22,172	$10,000	$0	-$12,172	-$82,742
15	Production	$14,830	$6,542	$800	$0	$22,172	$12,000	$0	-$10,172	-$92,914
16	Production	$14,830	$6,542	$800	$0	$22,172	$14,000	$0	-$8,172	-$101,086
17	Production	$14,830	$6,542	$800	$0	$22,172	$15,000	$0	-$7,172	-$108,258
18	Production	$14,830	$6,542	$800	$0	$22,172	$15,000	$0	-$7,172	-$115,430

Cash-out month: Month 9. Even with $75K investor capital arriving at Month 5, the cash runs out during Pilot 2.

Total capital required for 18 months: ~$190.4K ($75K founder + $75K investor + ~$40.4K shortfall).

With $100K founder capital: Cash-out shifts to Month 10. Still requires either revenue acceleration, scope reduction, or a second raise.

2.6 Cash Flow Summary Across Three Representative Paths

Metric	S1-L2	S2-O2-I2	S3-O2-I2
Founder capital	$200,000	$100,000	$75,000
Investor capital	$0	$75,000	$75,000
Total capital	$200,000	$175,000	$150,000
Cash-out month	Month 12	Month 7 (pre-investor)	Month 9
Total capital needed (full timeline)	~$336,500	~$304,700	~$265,400
Capital gap	~$136,500	~$54,700	~$40,400
Monthly burn at peak	$22,172	$22,172	$22,172
Months to potential breakeven	24-28	22-26	20-24
Revenue at breakeven	~$22,000/mo	~$22,000/mo	~$22,000/mo

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3. Runway Analysis

3.1 Investment Level Cost Multipliers

Investment Level	Team Cost Multiplier	Ops Staff	Infra	Monthly Burn (Pre-Pilot)	Monthly Burn (Pilot 1)	Monthly Burn (Pilot 2)	Monthly Burn (Production)
L1 (Bootstrapped, 40%)	0.40	Minimum of range	Minimum of range	$6,027	$8,621	$10,593	$12,992

L2 (Ideal, 100%)	1.00	Midpoint of range	Midpoint of range	$14,895	$17,582	$19,651	$22,172
L3 (Fully Funded, 175%)	1.75	Maximum of range	Maximum of range	$26,103	$31,467	$35,468	$40,858

L1 calculation detail:

- Team: $14,830 x 0.40 = $5,932/mo
- Ops staff: $0 / $1,440 / $2,612 / $3,840 (minimum of ranges, fewer staff, lower salaries)
- Infra: $65 / $160 / $300 / $600 (minimum of ranges per params.md Section 7.4)
- One-time costs: $200 / $400 / $300 (minimum of ranges per params.md Section 7.3)

L3 calculation detail:

- Team: $14,830 x 1.75 = $25,953/mo
- Ops staff: $0 / $3,524 / $6,110 / $10,458 (maximum of ranges, full staffing per params.md Section 4.2)
- Infra: $150 / $520 / $920 / $1,500 (maximum of ranges per params.md Section 7.4)
- One-time costs: $400 / $600 / $600 (maximum of ranges per params.md Section 7.3)

3.2 Assumed Starting Capital by Combination

Scenario	Our Level	Assumed Founder Capital	Investor Capital	Investor Timing	Total Available
S1-L1	L1	$80,000	$0	Never	$80,000
S1-L2	L2	$200,000	$0	Never	$200,000
S1-L3	L3	$400,000	$0	Never	$400,000
S2-O1-Ix	L1	$50,000	Varies	Transition	$50K + investor
S2-O2-Ix	L2	$100,000	Varies	Transition	$100K + investor
S2-O3-Ix	L3	$200,000	Varies	Transition	$200K + investor
S3-O1-Ix	L1	$30,000	Varies	After Pilot 1	$30K + investor
S3-O2-Ix	L2	$75,000	Varies	After Pilot 1	$75K + investor
S3-O3-Ix	L3	$150,000	Varies	After Pilot 1	$150K + investor

Investor capital by level:

Investor Level	Capital Provided
I1 (Bootstrapped)	$30,000-$40,000
I2 (Ideal)	$75,000-$100,000
I3 (Fully Funded)	$150,000-$250,000

3.3 Runway at Key Checkpoints — All 21 Combinations

Checkpoint definitions:

- CP1: End of Pre-Pilot (Month 4-6 depending on path)
- CP2: End of Pilot 1 (Month 6-9 depending on path)
- CP3: End of Pilot 2 (Month 11-18 depending on path)
- CP4: 6 months into Production

#	Code	Runway at CP1	Runway at CP2	Runway at CP3	Runway at CP4	Status
1	S1-L1	9.2 mo	5.8 mo	1.2 mo	DEPLETED	CRITICAL
2	S1-L2	9.4 mo	5.0 mo	DEPLETED	DEPLETED	CRITICAL
3	S1-L3	9.6 mo	5.2 mo	DEPLETED	DEPLETED	CRITICAL
4	S2-O1-I1	4.1 mo	1.2 mo + $35K inv = 5.1 mo	3.8 mo	1.2 mo	AT RISK
5	S2-O1-I2	4.1 mo	1.2 mo + $85K inv = 11.0 mo	6.5 mo	3.8 mo	VIABLE
6	S2-O1-I3	4.1 mo	1.2 mo + $200K inv = 24.0 mo	14.2 mo	9.5 mo	STRONG
7	S2-O2-I1	6.7 mo	2.5 mo + $35K inv = 4.3 mo	2.1 mo	DEPLETED	AT RISK
8	S2-O2-I2	6.7 mo	2.5 mo + $85K inv = 7.0 mo	3.5 mo	1.4 mo	VIABLE
9	S2-O2-I3	6.7 mo	2.5 mo + $200K inv = 12.8 mo	8.6 mo	5.2 mo	STRONG
10	S2-O3-I1	5.1 mo	1.8 mo + $35K inv = 2.8 mo	DEPLETED	DEPLETED	CRITICAL
11	S2-O3-I2	5.1 mo	1.8 mo + $85K inv = 4.5 mo	2.0 mo	DEPLETED	AT RISK
12	S2-O3-I3	5.1 mo	1.8 mo + $200K inv = 7.5 mo	3.8 mo	1.5 mo	VIABLE
13	S3-O1-I1	2.5 mo + $35K inv = 6.5 mo	4.8 mo	2.5 mo	DEPLETED	AT RISK
14	S3-O1-I2	2.5 mo + $85K inv = 12.3 mo	9.8 mo	5.5 mo	2.8 mo	VIABLE

15	S3-O1-I3	2.5 mo + $200K inv = 25.6 mo	20.5 mo	14.0 mo	9.2 mo	STRONG
16	S3-O2-I1	3.4 mo + $35K inv = 5.2 mo	3.8 mo	1.5 mo	DEPLETED	AT RISK
17	S3-O2-I2	3.4 mo + $85K inv = 7.7 mo	5.5 mo	2.8 mo	0.8 mo	AT RISK
18	S3-O2-I3	3.4 mo + $200K inv = 13.6 mo	10.2 mo	6.0 mo	3.2 mo	VIABLE
19	S3-O3-I1	2.6 mo + $35K inv = 3.5 mo	2.0 mo	DEPLETED	DEPLETED	CRITICAL
20	S3-O3-I2	2.6 mo + $85K inv = 5.0 mo	3.2 mo	1.0 mo	DEPLETED	AT RISK
21	S3-O3-I3	2.6 mo + $200K inv = 7.8 mo	5.2 mo	2.5 mo	0.5 mo	AT RISK

Status definitions:

- STRONG: Runway > 6 months at all checkpoints through CP4
- VIABLE: Runway > 3 months at most checkpoints; manageable with revenue growth
- AT RISK: Runway drops below 3 months at one or more checkpoints; requires action
- CRITICAL: Runway depletes before reaching Production; path is not viable without additional capital

3.4 Key Observations

1. **No self-funded path (S1) reaches Production without additional capital** at modeled burn rates and revenue assumptions. S1-L1 survives longest in calendar time due to lowest burn but produces the weakest product and growth.
2. **High own-spend levels (O3) paired with low investor levels (I1) are worse than moderate paths.** S2-O3-I1 and S3-O3-I1 burn through founder capital quickly and the small investor injection does not compensate.
3. **The optimal capital efficiency paths are S2-O1-I2, S2-O2-I2, S3-O1-I2, and S3-O2-I2** — moderate own spending paired with ideal investor capital.
4. **Only paths with I3 (fully funded investor) provide comfortable runway through Production.** All other paths require either revenue to materialize faster than projected, a second raise, or contingency actions.

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4. Cash-Out Risk Points

4.1 Definition

A **cash-out risk point** is the month at which cumulative cash position reaches $0 assuming no new capital injection beyond the initial plan and revenue follows the midpoint projection from params.md Section 11.3.

4.2 Cash-Out Month by Combination

#	Code	Founder Capital	Investor Capital	Investor Month	Cash-Out Month	Months of Operation
1	S1-L1	$80,000	$0	—	Month 13	13
2	S1-L2	$200,000	$0	—	Month 12	12
3	S1-L3	$400,000	$0	—	Month 12	12
4	S2-O1-I1	$50,000	$35,000	8	Month 14	14
5	S2-O1-I2	$50,000	$85,000	8	Month 19	19
6	S2-O1-I3	$50,000	$200,000	8	Month 32+	32+
7	S2-O2-I1	$100,000	$35,000	8	Month 12	12
8	S2-O2-I2	$100,000	$85,000	8	Month 14	14
9	S2-O2-I3	$100,000	$200,000	8	Month 22	22
10	S2-O3-I1	$200,000	$35,000	8	Month 10	10
11	S2-O3-I2	$200,000	$85,000	8	Month 12	12
12	S2-O3-I3	$200,000	$200,000	8	Month 16	16
13	S3-O1-I1	$30,000	$35,000	5	Month 12	12
14	S3-O1-I2	$30,000	$85,000	5	Month 18	18
15	S3-O1-I3	$30,000	$200,000	5	Month 30+	30+
16	S3-O2-I1	$75,000	$35,000	5	Month 10	10
17	S3-O2-I2	$75,000	$85,000	5	Month 13	13
18	S3-O2-I3	$75,000	$200,000	5	Month 20	20
19	S3-O3-I1	$150,000	$35,000	5	Month 9	9
20	S3-O3-I2	$150,000	$85,000	5	Month 11	11
21	S3-O3-I3	$150,000	$200,000	5	Month 15	15

4.3 Cash-Out Risk Heatmap

	I1 (Bootstrapped Investor)	I2 (Ideal Investor)	I3 (Fully Funded Investor)
S1 (No Investor)	L1: Mo 13; L2: Mo 12; L3: Mo 12	N/A	N/A
S2-O1 (Bootstrap Own)	Mo 14	Mo 19	Mo 32+
S2-O2 (Ideal Own)	Mo 12	Mo 14	Mo 22
S2-O3 (Full Own)	Mo 10	Mo 12	Mo 16
S3-O1 (Bootstrap Own)	Mo 12	Mo 18	Mo 30+
S3-O2 (Ideal Own)	Mo 10	Mo 13	Mo 20
S3-O3 (Full Own)	Mo 9	Mo 11	Mo 15

Pattern: Higher own-spending levels burn cash faster and shorten runway even when investor capital is constant. The most capital-efficient paths are those with modest own spending (O1) paired with strong investor capital (I2 or I3).

4.4 Paths That Survive to Production (Month 16+)

Only 7 of 21 combinations reach Month 16+ before cash-out:

#	Code	Cash-Out Month	Reaches Production?	Margin
5	S2-O1-I2	Month 19	Yes	3 months buffer
6	S2-O1-I3	Month 32+	Yes	16+ months buffer
9	S2-O2-I3	Month 22	Yes	6 months buffer
12	S2-O3-I3	Month 16	Barely	0 months buffer
14	S3-O1-I2	Month 18	Yes	2 months buffer
15	S3-O1-I3	Month 30+	Yes	14+ months buffer
18	S3-O2-I3	Month 20	Yes	4 months buffer

Conclusion: Reaching Production requires either (a) fully funded investor capital (I3 = $150K-$250K), or (b) bootstrapped own spending with ideal investor capital (O1-I2), or (c) revenue exceeding projections by 50%+, or (d) a second funding round.

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5. Minimum Cash Reserve Thresholds

5.1 Threshold Framework

Threshold Level	Remaining Runway	Cash Position Trigger	Status
GREEN	> 6 months	> 6x monthly burn	Normal operations
YELLOW	4-6 months	4-6x monthly burn	Begin contingency planning
ORANGE	2-4 months	2-4x monthly burn	Execute contingency actions (see Section 7)
RED	< 2 months	< 2x monthly burn	Emergency mode; survival actions only

5.2 Threshold Values by Phase

Phase	Monthly Burn (L2)	GREEN Threshold	YELLOW Threshold	ORANGE Threshold	RED Threshold
Pre-Pilot	$14,895	> $89,370	$59,580-$89,370	$29,790-$59,580	< $29,790
Pilot 1	$17,582	> $105,492	$70,328-$105,492	$35,164-$70,328	< $35,164
Pilot 2	$19,651	> $117,906	$78,604-$117,906	$39,302-$78,604	< $39,302
Production	$22,172	> $133,032	$88,688-$133,032	$44,344-$88,688	< $44,344

5.3 Threshold Values by Investment Level

Level	Peak Monthly Burn	RED Threshold (< 2 months)
L1	$12,992 (Production)	< $25,984
L2	$22,172 (Production)	< $44,344
L3	$40,858 (Production)	< $81,716

5.4 Monitoring Protocol

Frequency	Action	Responsible
Weekly	Review cash position against thresholds	Founder
Monthly	Update cash flow projections with actuals	Founding Member 1
Quarterly	Full runway review with scenario re-modeling	Founder + Advisors

On threshold change	Immediate notification to all founders and advisors	Automated alert

5.5 Threshold Transition Actions

Transition	Immediate Action
GREEN to YELLOW	Begin identifying contingency levers; update investor materials; start informal investor conversations if S2/S3 path
YELLOW to ORANGE	Activate contingency plan (Section 7); reduce non-essential spend; accelerate revenue initiatives; begin active fundraising
ORANGE to RED	Emergency mode: freeze all non-critical hiring; cut infra to minimum; defer all non-essential development; pursue bridge financing or revenue advance
Any level to GREEN	Resume normal operations; document lessons learned; adjust projections

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6. Investor Impact on Runway

6.1 Runway Extension Formula

For each investor dollar received, the runway extension depends on the current burn rate:

Monthly Burn	Runway Extension per $10K Invested	Per $50K	Per $100K	Per $200K
$8,700 (L1, Pilot 1)	1.15 months	5.7 mo	11.5 mo	23.0 mo
$17,600 (L2, Pilot 1)	0.57 months	2.8 mo	5.7 mo	11.4 mo
$19,650 (L2, Pilot 2)	0.51 months	2.5 mo	5.1 mo	10.2 mo
$22,170 (L2, Production)	0.45 months	2.3 mo	4.5 mo	9.0 mo
$31,500 (L3, Pilot 1)	0.32 months	1.6 mo	3.2 mo	6.3 mo
$40,860 (L3, Production)	0.24 months	1.2 mo	2.4 mo	4.9 mo

Key insight: Investor capital is most impactful when the burn rate is low. A $75K investment at L1 burn extends runway by 8.6 months; the same $75K at L3 burn extends runway by only 1.8 months at Production.

6.2 Scenario S2: Investor During Pilot 1-to-Pilot 2 Transition

Context: The investor arrives when Health Hub has completed Pilot 1 and demonstrated initial traction (up to 1,000 users, first revenue, consultation completion data). Equity given: 15-20% (per params.md Section 8.3).

Own Level	Investor Level	Investor Capital	Runway Before Investor	Runway After Investor (additional months)	Total Remaining Runway
O1	I1	$35,000	1.2 mo	+4.1 mo	5.3 mo
O1	I2	$85,000	1.2 mo	+9.9 mo	11.1 mo
O1	I3	$200,000	1.2 mo	+23.2 mo	24.4 mo
O2	I1	$35,000	2.5 mo	+2.0 mo	4.5 mo
O2	I2	$85,000	2.5 mo	+4.8 mo	7.3 mo
O2	I3	$200,000	2.5 mo	+11.4 mo	13.9 mo
O3	I1	$35,000	1.8 mo	+1.1 mo	2.9 mo
O3	I2	$85,000	1.8 mo	+2.7 mo	4.5 mo
O3	I3	$200,000	1.8 mo	+6.4 mo	8.2 mo

6.3 Scenario S3: Investor After Pilot 1

Context: The investor arrives earlier (right after Pilot 1), which means less traction data but earlier capital. The earlier arrival means the burn rate at the time of investment is lower (still in Pilot 1 / early Pilot 2 transition).

Own Level	Investor Level	Investor Capital	Runway Before Investor	Runway After Investor (additional months)	Total Remaining Runway
O1	I1	$35,000	2.5 mo	+4.1 mo	6.6 mo
O1	I2	$85,000	2.5 mo	+9.9 mo	12.4 mo
O1	I3	$200,000	2.5 mo	+23.2 mo	25.7 mo
O2	I1	$35,000	3.4 mo	+2.0 mo	5.4 mo
O2	I2	$85,000	3.4 mo	+4.8 mo	8.2 mo
O2	I3	$200,000	3.4 mo	+11.4 mo	14.8 mo
O3	I1	$35,000	2.6 mo	+1.1 mo	3.7 mo
O3	I2	$85,000	2.6 mo	+2.7 mo	5.3 mo
O3	I3	$200,000	2.6 mo	+6.4 mo	9.0 mo

6.4 Investor Timing Sensitivity

Timing Factor	S2 (During Transition)	S3 (After Pilot 1)	Difference
Month of arrival	Month 7-8	Month 5-6	S3 is 2-3 months earlier
Traction data available	Pilot 1 complete; 1,000 users	Pilot 1 in-progress; 300-500 users	S2 has stronger data
Burn rate at arrival	$17,582-$19,651/mo	$14,895-$17,582/mo	S3 burn is lower = capital goes further
Equity expected	15-20%	18-20%	S3 may require more equity for same capital due to less traction
Fundraising difficulty	Moderate (traction proven)	Higher (traction unproven)	S2 is easier to close

Recommendation: S2 timing is preferred when investor capital > $75K because the stronger traction data makes the raise easier to close. S3 timing is preferred when the founder's self-funded runway is < 5 months at Pilot 1 start, making an earlier raise a survival necessity.

6.5 Minimum Viable Investor Capital

What is the minimum investor capital that makes each path viable (defined as reaching at least Month 16 with positive cash)?

Path	Minimum Investor Capital	Context
S2-O1	$65,000	Modest own spend; investor covers Pilot 2 + early Production
S2-O2	$130,000	Ideal own spend; investor must cover the higher burn differential
S2-O3	$200,000+	High own spend burns through capital quickly; even large raises barely suffice
S3-O1	$55,000	Earlier arrival means capital stretches further
S3-O2	$120,000	Similar to S2 but slightly less needed due to earlier timing
S3-O3	$180,000+	High own spend is problematic regardless of investor timing

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7. Contingency Actions

7.1 Contingency Action Framework

Contingency actions are organized by severity level and by the type of lever being pulled. The goal is to extend runway while preserving maximum optionality.

7.2 Tier 1: Scope Reduction (YELLOW threshold, 4-6 months runway)

Action	Monthly Savings	Impact	Reversibility
Defer Kenya launch preparation	$500-$1,000	Delays second-market entry by 3-6 months; Ethiopia focus remains	High — can restart at any time
Reduce Android team from 4 to 2 developers	$1,760-$2,640	APK timeline extends by 2-3 months; core features still delivered	Medium — rehiring takes 2-4 weeks
Pause specialist consultation feature development	$1,000-$1,500	Limits platform to GP consults only in near term	High — code is preserved
Defer analytics/reporting dashboard work	$800-$1,200	Manual reporting continues; no self-serve investor dashboards	High
Reduce advisory board engagement to quarterly	$500-$750	Fewer strategic inputs; acceptable during execution phases	High

Total Tier 1 savings: $4,560-$7,090/mo (reduces L2 burn from ~$17,600 to ~$10,500-$13,000 during Pilot 1)

7.3 Tier 2: Hiring Deferral (ORANGE threshold, 2-4 months runway)

Action	Monthly Savings	Impact	Reversibility
Defer ops staff hiring by 1 phase	$2,482-$4,361	Service coverage reduced; longer patient wait times; manual callback load on founders	Medium — hiring takes 2-4 weeks in India
Reduce doctor coverage from 2-shift to 1-shift	$720	Coverage drops from ~16 hrs/day to ~8 hrs/day; limits service to daytime only	Medium
Defer tech support hiring; founders absorb L1 support	$301-$602	Founder time diverted from strategic work; acceptable short-term	High
Reduce infra to bare minimum tier	$100-$300	Performance degradation; acceptable for < 500 active users	High
Pause all non-critical one-time equipment purchases	$0-$500 (amortized)	Use personal devices temporarily	High

Total Tier 2 savings: $3,603-$5,963/mo (combined with Tier 1: burn can drop to ~$5,000-$8,000/mo)

7.4 Tier 3: Bridge Financing (ORANGE to RED threshold)

Action	Amount	Timeline to Close	Terms
Founder bridge loan	$10,000-$30,000	Immediate	Interest-free; converted to equity at next round's valuation

Friends & family note	$10,000-$50,000	2-4 weeks	Convertible note; 20% discount at next round
Revenue advance / pre-sale	$5,000-$15,000	2-6 weeks	Sell annual facility subscriptions at discount; pre-sell consultation packages
Grant / competition prize	$5,000-$25,000	4-12 weeks	Non-dilutive; many East African health tech grant programs available
Strategic partner advance	$10,000-$30,000	4-8 weeks	Clinic chain or pharmacy network pays upfront for platform access

7.5 Tier 4: Emergency Survival (RED threshold, < 2 months runway)

Action	Monthly Savings	Impact	Reversibility
Reduce team to founders only (3 people)	$9,550	All development pauses; maintenance mode only; ops continue manually	Low — team dissolution is hard to reverse
Convert all team payments to deferred equity	$14,830	Cash burn drops to ops + infra only; team works on equity promise	Low — risky; team may leave
Pause all ops staff; founders handle callbacks manually	$2,482-$6,542	Service quality drops dramatically; only viable with < 200 active users	Medium
Shut down non-essential infrastructure	$200-$600	Reduced to single server; video consultations may be unavailable	Medium
Seek acqui-hire or strategic merger	N/A	Company survives as part of larger entity; founder loses majority control	Irreversible

Total Tier 4 savings: up to $25,000/mo (reduces burn to $2,000-$5,000/mo — survival mode only)

7.6 Contingency Decision Matrix

Cash Runway	First Action	If Still Declining	If Stabilized
6 months	Monitor weekly	Begin Tier 1 scope cuts	Maintain course
5 months	Tier 1 scope cuts	Start fundraising conversations	Reassess in 30 days
4 months	Active fundraising + Tier 1 fully executed	Add Tier 2 hiring deferrals	Continue fundraising
3 months	Tier 1 + Tier 2 fully executed	Pursue Tier 3 bridge financing	Close bridge or round
2 months	Tier 3 bridge executed or in progress	Begin Tier 4 emergency actions	Bridge must close within 30 days
1 month	Tier 4 fully executed	Seek acqui-hire or orderly wind-down	Any capital injection buys time

7.7 Contingency Impact on Metrics

Contingency Tier	Burn Reduction	Timeline Impact	Metric Impact
Tier 1 (Scope)	-$4.5K-$7K/mo	+2-3 months to milestones	Lower MAU growth; delayed specialist features; no Kenya prep
Tier 2 (Hiring)	-$3.6K-$6K/mo	+1-2 months to milestones	Reduced consultation completion rate; lower doctor utilization; higher response times
Tier 3 (Bridge)	N/A (adds capital)	No delay if closed quickly	Potential dilution or debt obligation; minor distraction
Tier 4 (Emergency)	-$10K-$25K/mo	Indefinite delay	All growth metrics frozen; product stagnation; high risk of user and team attrition

7.8 Pre-Positioned Contingency Preparations

Actions to take during GREEN status to prepare for potential future downturns:

Preparation	When	Cost	Purpose
Maintain updated pitch deck	Ongoing	Time only	Ready for opportunistic fundraising
Build relationships with 5-10 potential investors	Pre-Pilot through Pilot 1	Time only	Pipeline ready if fundraising needed
Identify and document potential grant programs	Pre-Pilot	Time only	Non-dilutive capital options mapped
Negotiate deferred-payment terms with key vendors	Pilot 1	Time only	Reduce immediate cash outflow in crunch
Create a "minimum viable service" configuration	Pre-Pilot	20-30 hours	Know exactly what to cut and what to keep
Pre-negotiate convertible note terms with advisors	Pre-Pilot	Legal review $500-$1,000	Bridge financing ready to activate quickly

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8. Cross-References

Topic	Document
Foundation parameters and assumptions	params.md
Business metrics definitions and targets	[Document 09 — Business Metrics]()
Cost breakdown by phase and scenario	Document 03 — Cost Breakdown

Revenue modelling across 21 combinations	Document 05 — Revenue Modelling
Effort estimation and timeline	Document 04 — Effort Estimation
Timeline and phase definitions	Document 01 — Timeline
Equity and capital structure	params.md Section 8

End of Document 10 — Cash Flow & Runway. All figures trace to params.md v2.